Archie Qian

Toronto ON | +1(825) 984 6233 | Data Scientist | archibaldchain@gmail.com | LinkedIn

Summary

- Machine learning engineer with 5+ years of experience developing predictive analytics and data pipelines across industrial infrastructure, energy systems, and financial datasets.
- Advanced Python expertise for machine learning, time-series forecasting, and scalable data processing.
- Experienced in building ETL pipelines and working with large datasets using SQL systems including PostgreSQL, SQL Server, and Microsoft Access.
- Applied machine learning and optimization models to infrastructure problems including electricity demand forecasting, energy system optimization, and asset risk modeling.

WORK EXPERIENCE

BBA Engineer Inc. — *Calgary, AB – Toronto, ON*

Industrial Systems Data Analytics — *Sept. 2024 – Present*

- Developed an **hourly battery dispatch optimization model** using Linear Programming in Python to simulate energy storage operations based on long-term electricity price forecasts spanning 20+ years.
- Built **time-series forecasting models** (ARIMA, regression) to predict electricity load from historical operational data, achieving <12% MAPE and supporting power system planning analysis.
- Conducted **A/B experiments on EV charging load profiles** across Canadian regions, demonstrating that managed charging strategies reduce peak demand impact by ~20%.
- Built **predictive risk and failure probability models** for 100,000+ power transmission assets using operational and maintenance datasets, supporting infrastructure asset management and guiding $1M+ in maintenance planning.
- Designed **Python-based ETL pipelines** integrating large-scale infrastructure asset data from SQL Server into automated analytics dashboards (Power BI) for asset lifecycle monitoring and operational insights.
- Performed **cybersecurity risk assessments for industrial control systems** including SCADA environments and OT device inventories, supporting NERC CIP compliance and improving risk visibility across utility infrastructure systems.

Intact Financial Corporation — *Calgary, AB*

Data Scientist Intern — *Oct. 2023 – Jan. 2024*

- Developed **NLP classification models** using Large Language Models and Sentence Transformers (PyTorch) to categorize call transcripts into 10+ operational categories, achieving >60% F1 score.
- Built and deployed **ML pipelines in AWS SageMaker**, integrating transcript data from S3 and monitoring model performance using MLflow.
- Designed **ETL workflows using Python and SQL** to process large-scale call transcript datasets and enable automated analytics for customer service insights.
- Implemented **rule-based Named Entity Recognition (NER)** using Python and regular expressions to extract key entities from call transcripts, improving downstream business analytics.

Cenozon Inc. — *Calgary, AB*

Data Scientist Intern — *July. 2023 – Oct. 2023*

- Analyzed oil pipeline operational datasets including production rates, fluid composition, and pipeline network metrics using SQL and Python.
- Developed machine learning regression models (Linear Regression, Decision Tree, Neural Networks) to predict pipeline corrosion rates and infrastructure risk, achieving >90% accuracy relative to engineering baseline calculations.
- Built data processing pipelines to integrate Excel-based inspection readings with production databases, improving data completeness and enabling automated analysis workflows.

RESEARCH EXPERIENCE

Machine Learning Researcher | Cross-validation Correction — *Sept. 2021 – Sept. 2023*

- Developed a statistical structure using covariance between the Training set and the Test set to adjust the Cross-Validation Error.
- Explained and tested corrected CV error for the models of Linear Mixed-Effects Model (LMM), Best Linear Unbiased Predictor (BLUP), Ridge regression, and Bayesian Selection Linear Mixed Model (BSLMM).
- Simulated large dataset from 1000 Genomes Project (60 Gb) using cloud computing (Advanced Research Computing) with Python frameworks (Numpy, Pandas, and Scikit-learn), which has a closer estimation of generalization error than CV error.

- Published: The Bias of Using Cross-Validation in Genomic Predictions and Its Correction

Machine Learning Engineer | Reinforcement Learning in Robotic Arm *Oct. 2022 – Apr. 2023*

- Collaborated with a cross-functional team of six to deliver the project within the set timeline and scope.
- Worked as part of the Robotic Arm Algorithm Team, taking ownership of the end-to-end reinforcement learning pipeline for training the Kinova Gen3 Arm to achieve object manipulation tasks, achieving a success rate of over 90%.
- Implemented the DDPG algorithm using PyTorch to control the 8 motors of the robot and its 7 degrees of freedom, leveraging MuJoco and OpenAI Gym for simulation and testing.
- Transferred the learned policy to the physical robot, enabling it to successfully reach, fetch, and pick-and-place objects using Computer Vision through OpenCV.

EDUCATION

MSc in Statistics (GPA: 3.9/4) University of Calgary *Calgary, Canada*

Thesis: Cross-Validation Correction of Machine Learning Algorithms in Genomic Data *Sept. 2021 – Sept. 2023*

BSc in Applied Mathematics (GPA: 3.7/4) *Shenzhen, China*

Southern University of Science and Technology

SKILLS

Programming: Python (NumPy, Pandas, Scikit-Learn, PyTorch, TensorFlow, XGBoost), R, SQL, Java, C++, JavaScript

Tools: Git, PowerBI, AWS(S3, Sagemaker, Bedrock, Lambda, API Gateway), Azure Databrick, MLflow, Snowflakes, Docker, Hadoop

Analytical Tools: Data Mining, Machine Learning, Data Pipeline, LLMs, A/B Test, Google Analytics